LLaMA-Unidetector: An LLaMA-Based Universal Framework for Open-Vocabulary Object Detection in Remote Sensing Imagery

Guanqun Wang, *Member, IEEE*, Jianlin Xie, Tong Zhang, *Graduate Student Member, IEEE*, Yikang Sun, He Chen, Yin Zhuang, *Member, IEEE*, and Jun Li, *Fellow, IEEE*

***Abstract*— The object detection is a crucial task in the computer vision for remote sensing applications. However, the reliance of traditional methods on predefined and trained object categories limits their applicability in open-world scenarios. A key challenge in open-vocabulary object detection lies in accurately identifying unseen objects. Existing approaches often focus solely on detecting object locations, struggling to recognize the categories of previously unseen targets. To address this issue, we propose a novel benchmark, where models are trained on known base classes and evaluated on their performance in detecting and recognizing unseen or novel classes. To this end, we introduce LLaMA-Unidetector, a universal framework that incorporates textual information into a closed-set detector, enabling the generalization to open-set scenarios. Our LLaMA-Unidetector leverages a decoupled learning strategy that separates localization and recognition. In the first stage, a class-agnostic detector identifies objects, distinguishing only between foreground and background. In the second stage, the detected foreground objects are passed through TerraOV-LLM, a multimodal large language model (MLLM), for recognition, utilizing the strong generalization capabilities of large language models to infer the correct categories. We propose a self-built vision question answering (VQA) remote sensing dataset, TerraVQA, and conduct extensive experiments on the NWPU-VHR10, DOTA1.0, and DIOR datasets. The LLaMA-Unidetector achieves impressive results, with a performance of 75.46% AP, 50.22% AP, and 51.38% AP on the zero-shot detection benchmarks for the NWPU-VHR10, DOTA1.0, and DIOR datasets, respectively. Our source code is available at: https://github.com/ChloeeGrace/LLaMA-Unidetector**

***Index Terms*— Decoupled learning, open vocabulary, remote sensing object detection.**

Received 13 January 2025; revised 27 March 2025; accepted 20 April 2025. Date of publication 25 April 2025; date of current version 8 May 2025. This work was supported in part by the General Program of National Natural Science Foundation of China under Grant 62371048, in part by the National Natural Science Foundation of China under Grant T2225019, and in part by the Beijing Institute of Technology (BIT) Research and Innovation Promoting Project under Grant 2024YCXY030. *(Jianlin Xie and Guanqun Wang contributed equally to this work.) (Corresponding author: Guanqun Wang.)*

Guanqun Wang is with the National Key Laboratory for Multimedia Information Processing, Peking University, Beijing 100871, China (e-mail: guanqun2023@gmail.com).

Jianlin Xie, Tong Zhang, Yikang Sun, He Chen, and Yin Zhuang are with the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing (SBIIP), Beijing Institute of Technology, Beijing 100081, China.

Jun Li is with the School of Computer Science and Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China.

Digital Object Identifier 10.1109/TGRS.2025.3564332

I. Introduction

OBJECT detection in remote sensing imagery plays a crucial role in remote sensing AI interpretation, aiming to locate and classify valuable objects, such as aircraft, ships, and vehicles within images. As a fundamental and important task, remote sensing object detection has widespread applications in large-scale environmental monitoring [1], [2], [3], traffic guidance [4], [5], object tracking [6], and urban planning [7], [8], [9].

The current research in the object detection predominantly focuses on closed-set detection, where detectors are trained on datasets with predefined categories, such as the COCO [10] and Objects365 [11] datasets. These detectors are then used to identify objects within the fixed set of categories. However, this approach is limited by the virtually infinite pool of concepts present in real-world remote sensing scenarios, where objects may fall outside the predefined categories. As illustrated in Fig. 1, this challenge is known as open-vocabulary object detection. Recent studies [12], [13], [14] integrating the hyperspectral data have provided additional spectral information, further enhancing the capability of intelligent interpretation of remote sensing imagery. In the practical remote sensing image interpretation, the appearance of previously unseen or noncooperative targets necessitates the development of methods for detecting these new targets. Thus, detecting noncooperative targets in remote sensing imagery is an urgent task that will become a new paradigm for future object detection.

While many detectors [15], [16], [17], [18], [19], [20], [21], [22] have achieved commendable performance on conventional remote sensing datasets, their reliance on predefined and trained object categories restricts their applicability in open-world scenarios. For example, a model trained to detect aircraft may not be suitable for detecting vehicles. In addition, existing methods heavily depend on manually annotated data. When new object categories need to be added or modified, it requires reannotating the training data and retraining the model, which is both time-consuming and labor-intensive. Furthermore, real-world scenarios often present an almost limitless number of unknown object categories as new objects continuously emerge and evolve. This raises a challenging question: how can we detect a vast number of unknown categories without incurring the cost of extensive supervision?

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

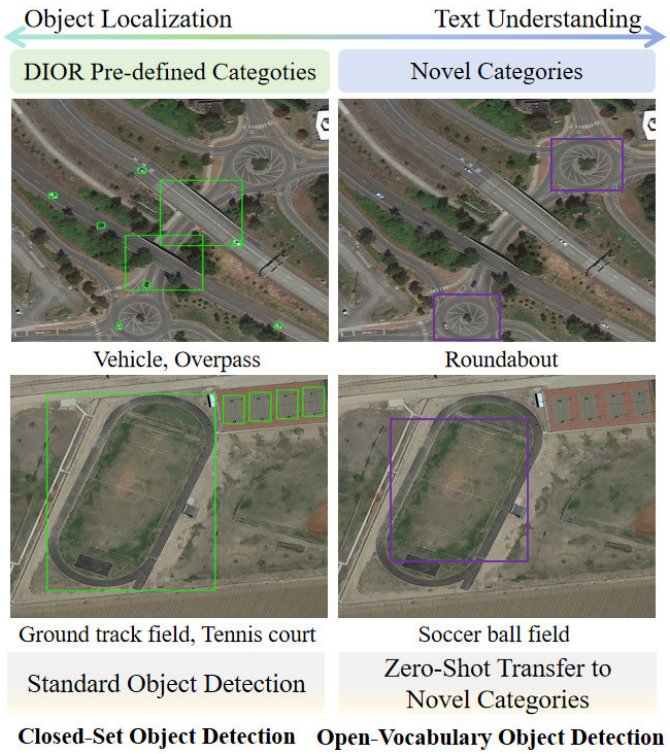


Fig. 1. Open-vocabulary object detection. Closed-set object detection requires models to detect targets belonging to predefined categories in the dataset, such as vehicles, overpass, GTF, and TC. In contrast, the open-vocabulary object detection incorporates text comprehension, leveraging textual information to enable the model to generalize by transferring from base classes to novel categories.

To address this issue, researchers have carried out a series of studies on open-vocabulary object detection in recent years, which can be categorized into three strategies: knowledge distillation [Fig. 2(a)], region-text alignment [Fig. 2(b)], and region-text pretraining [Fig. 2(c)]. The goal of knowledge distillation is to directly distill visual knowledge aligned with text into the closed-set detectors. Gu et al. [23] proposed ViLD, a training method that distills knowledge from a pretrained open-vocabulary image classification model (teacher) into a two-stage detector (student) through vision and language knowledge distillation. Wang et al. [24] proposed an object-aware distillation pyramid (OADP) framework to address the limitations of nonadaptive proposal cropping and single-level feature mimicking, which cause information destruction and inefficient knowledge transfer, by incorporating an object-aware knowledge extraction (OAKE) module and a distillation pyramid (DP) mechanism. Ma et al. [25] proposed a hierarchical visual-language knowledge distillation (HierKD) method for open-vocabulary one-stage detection, which combines global-level knowledge distillation to transfer knowledge of unseen categories from the pretrained visual-language model (PVLM) to the detector with instance-level knowledge distillation to simultaneously learn knowledge of seen and unseen categories. Cho et al. [26] proposed a simple and effective method named pseudo-caption labeling (PCL), which argues for the necessity of fine-grained labels to extract richer knowledge about novel objects, including their attributes and relationships, by using an image captioning model to generate captions that describe object instances from diverse perspectives, resulting in dense samples for knowledge distillation. However, during the distillation process, some fine-grained information may be lost, leading to a decrease in the student model's performance when handling complex scenes and details. Region-text alignment leverages large-scale, easily accessible region-text data to map the visual and textual features of new classes into an aligned feature space. Zareian et al. [27] proposed a new method, called OVR-convolutional neural network (CNN), to train object detectors using limited bounding box annotations and image–caption pairs, achieving significantly higher accuracy than zero-shot approaches for detecting and localizing objects without bounding box annotations during training. Zhong et al. [28] introduced RegionCLIP, a method that extends CLIP to learn region-level visual representations, enabling fine-grained alignment between image regions and textual concepts by using a CLIP model to match image regions with template captions and pretraining the model to align these region-text pairs in the feature space. However, the region-text pretraining method utilizes large scale, easily obtainable image–text pairs, whose textual data quality varies greatly. Low-quality or inaccurate text descriptions can affect the model's pretraining effectiveness and generalization ability. The goal of region-text alignment is to better match regional visual features with textual features during the training process. Wu et al. [29] proposed a method that aligns the embedding of a bag of regions beyond individual regions by grouping contextually interrelated regions and treating their embeddings as words in a sentence, which are then sent to the text encoder of a VLM to obtain the bag-of-regions embedding that is aligned to the corresponding features extracted by a frozen VLM. Zang et al. [30] proposed a novel open-vocabulary detector based on detection transformer, called OV-DETR, which formulates the learning objective as a binary matching task between input queries (class names or exemplar images) and corresponding objects, enabling the model to learn useful correspondences and generalize to unseen queries during testing. Kuo et al. [31] presented F-VLM, a simple open-vocabulary object detection method built upon frozen vision and language models, which simplifies the current multistage training pipeline by eliminating the need for knowledge distillation or detection-tailored pretraining. Lin et al. [32] proposed VLDet, a novel open-vocabulary object detection framework that learns directly from image–text pairs, simplifying training through set matching between image region features and word embeddings. Compared with the region-text pretraining method, the region-text alignment method requires a large number of accurately labeled image regions and corresponding text descriptions. This is difficult and expensive, increasing the cost of data collection and annotation.

Despite the advancements achieved by knowledge distillation, region-text alignment, and region-text pretraining, these methods share fundamental limitations that hinder their applicability in open-vocabulary object detection. Mathematically, the core optimization objectives of these methods can be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

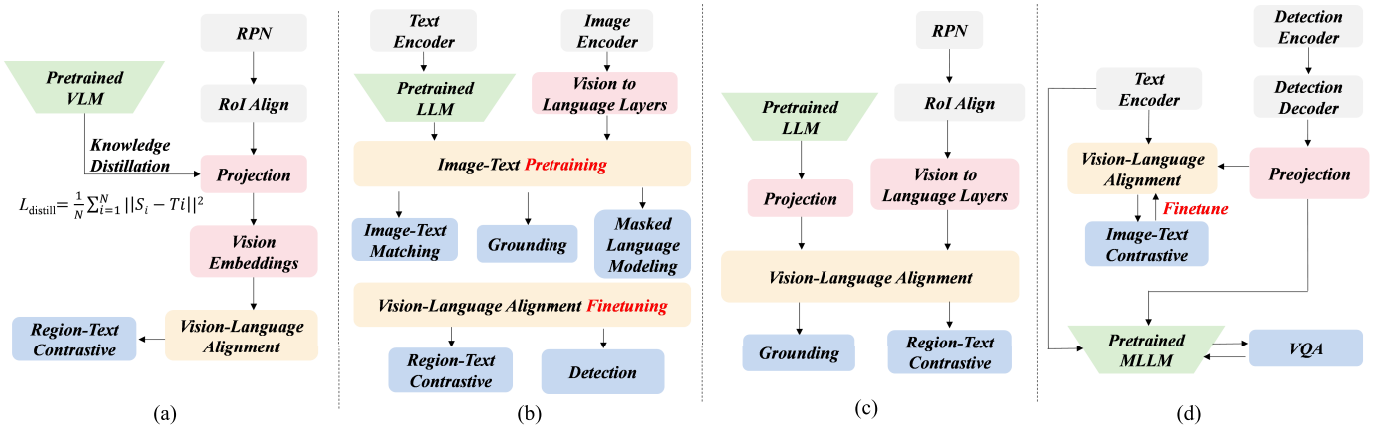


Fig. 2. Comparison of open-vocabulary object detection strategies. (a) Knowledge distillation. (b) Region-text alignment. (c) Region-text pretraining. (d) Our proposed method.

expressed as follows:

$$\mathcal{L}_{\text{existing}} = \mathcal{L}_{\text{align}} + \mathcal{L}_{\text{task-specific}} + \mathcal{L}_{\text{distill}} \tag{1}$$

where $\mathcal{L}_{\text{align}}$ aligns the visual ($\mathbf{v}_i$) and textual ($\mathbf{t}_i$) features, typically by minimizing the direct feature-level discrepancy, such as follows:

$$\mathcal{L}_{\text{align}} = \sum_{i=1}^{N} \|\mathbf{v}_i - \mathbf{t}_i\|^2. \tag{2}$$

$\mathcal{L}_{\text{task-specific}}$ focuses on optimizing task-level objectives like detection or classification, e.g.,

$$\mathcal{L}_{\text{task-specific}} = \sum_{i=1}^{N} \ell\left(y_i, \hat{y}_i\right) \tag{3}$$

and $\mathcal{L}_{\text{distill}}$ enforces knowledge transfer between teacher and student models via feature mimicking

$$\mathcal{L}_{\text{distill}} = \sum_{i=1}^{N} \left\|\mathbf{f}_i^{\text{teacher}} - \mathbf{f}_i^{\text{student}}\right\|^2. \tag{4}$$

While these formulations have enabled progress, they inherently rely on rigid feature mappings. Such dependence limits the ability of existing methods to generalize to novel or unseen categories, as the learned representations are constrained by the fixed semantic relationships captured during training. Furthermore, the coupled nature of detection and recognition tasks exacerbates this limitation by entangling spatial localization with semantic reasoning, making the optimization process highly dependent on task-specific data and introducing inefficiencies when adapting to new scenarios.

To overcome these limitations, we propose the LLaMA-Unidetector framework, which introduces a decoupled learning strategy that separates object localization and category recognition into two independent stages. In the first stage, a class-agnostic detection branch is designed to identify object regions by distinguishing foreground from background. Unlike traditional object detectors that rely on predefined categories, this branch employs a probabilistic modeling framework to enhance generalization capabilities in open-world scenarios. Our method involves two key components in this stage: a foreground–background discrimination task to classify regions as either foreground or background and a localization task to align predicted bounding boxes with ground truth annotations. Through foreground–background classification and region localization, the class-agnostic detection branch focuses on learning general spatial and contextual features, thereby providing a solid foundation for robust generalization in open-world scenarios. To further enhance performance, the class-agnostic detector incorporates probabilistic regularization and geometric optimization techniques to handle the inherent variability in object detection tasks.

By removing multiclass dependencies and explicitly modeling both the geometric and probabilistic aspects of object detection, the localization branch not only learns general spatial features but also achieves robust generalization across diverse scenarios and unseen object categories.

The second stage performs category recognition using TerraOV-LLM, a multimodal large language model (MLLM) that aligns detected visual features with semantic textual prompts in a shared latent space. Our approach leverages contrastive learning to dynamically align visual and textual representations. It explicitly encourages semantic alignment between modalities, enabling zero-shot and open-world generalization by leveraging the pretrained knowledge of the language model. In addition, to fully leverage the strengths of both image and text modalities and enhance the model's comprehension and reasoning capabilities, visual–textual multimodal fusion features are obtained by integrating textual and visual features through cross-modality attention, enabling complementary fusion of modality-specific information.

TerraOV-LLM may need to focus on finer-grained local features, whereas the detection bounding box provides only coarse-grained visual information. In addition, differing optimization objectives may lead the two stages to learn distinct representations, potentially affecting the overall task consistency. To ensure the coherence of visual features and mitigate inconsistencies in training objectives between the two stages, we introduce a feature alignment mechanism by a learnable projection matrix $\mathbf{W}_p$, which projects the output $\mathbf{f}_i$ of the localization branch into a compatible feature space for the recognition branch

$$\mathbf{f}_i' = \mathbf{W}_p \cdot \mathbf{f}_i. \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

This decoupled strategy provides a theoretically grounded solution to the limitations of existing methods. By separating spatial and semantic reasoning tasks, our framework reduces optimization complexity and enhances modularity, allowing independent improvements to each stage. The contrastive alignment objective ensures robust cross-modal semantic generalization, while the feature consistency mechanism bridges the gap between localization and recognition. Together, these innovations establish a scalable and adaptable approach to open-vocabulary object detection, addressing the challenges posed by unseen categories and dynamic real-world scenarios. Beyond the fixed datasets used, open-vocabulary object detection has broad practical applications in disaster monitoring. It can identify various types of postdisaster damage and affected areas, such as the extent of infrastructure destruction (e.g., buildings and roads), and even detect targets involved in postdisaster rescue efforts. Due to its open-vocabulary nature, the model can recognize new object categories beyond the training set, enabling it to adapt to the diverse and unforeseen scenarios in disaster-stricken areas. This provides accurate and timely monitoring data that supports postdisaster recovery and urban planning, thereby enhancing the efficiency and accuracy of postdisaster assessments.

Furthermore, we propose a new benchmark for remote sensing object detection. This benchmark utilizes several publicly available datasets with diverse categories, including NWPU-VHR10 [33], DOTA [34], and DIOR [8]. We design an open-set evaluation protocol to define the training and testing processes: during training, only fixed base classes are used, while testing evaluates the model's performance on unseen or novel classes. This benchmark offers a new evaluation approach for remote sensing object detection in open-world scenarios, providing a more realistic assessment of the model's performance when encountering unknown targets. Through this setup, we can comprehensively measure the model's generalization ability in an open-vocabulary environment.

In conclusion, the main contributions of this article are summarized as follows.

1) We introduce a novel open-set benchmark for remote sensing, accompanied by a self-built vision question answering (VQA) remote sensing dataset, TerraVQA, providing a platform for researchers to explore open-vocabulary object detection tasks in remote sensing imagery.
2) We propose a class-agnostic detector that eliminates multiclass dependencies while explicitly modeling both geometric and probabilistic aspects of object detection, enabling the localization branch to learn general spatial representations and achieve strong generalization across diverse scenarios and unseen object categories.
3) We propose an LLaMA-based MLLM without class vocabulary limitations that offers superior semantic understanding and the flexibility to handle novel concepts beyond the vocabulary.
4) We conduct extensive experiments on three publicly available remote sensing object detection datasets, demonstrating the effectiveness of our approach with significant performance improvements.

II. Related Works

In this section, some works related to object detection, open-vocabulary object detection, and vision–language models will be reviewed.

A. Object Detection

Object detection has been a prominent research focus in the field of computer vision, particularly with the rapid advancements in deep learning methods.

CNN-based object detection models have achieved remarkable success and significantly advanced the state of the art. Ren et al. [35] introduced Faster R-CNN, which integrates a region proposal network (RPN) with shared weights between the RPN and the detection network, reducing the computational burden of region proposal generation and thereby speeding up detection. Redmon [36], [38] and Redmon and Farhadi [37] proposed you only look once (YOLO), a framework that divides the image into regions and predicts bounding boxes and confidence scores for each. Sun et al. [39] developed sparse R-CNN, a purely sparse method that eliminates the need to enumerate object candidate locations across all dense image grids and avoids interacting with global dense image features. Tian et al. [40] proposed the fully convolutional one-stage (FCOS) object detector, which tackles object detection with per-pixel prediction in a manner similar to semantic segmentation. Transformer-based object detection models have also seen rapid development. Carion et al. [41] introduced DETR, an end-to-end model that removes the need for hand-crafted components like anchor design and NMS. To address the slow convergence and limited feature resolution issues of DETR due to the Transformer's attention mechanism, Zhu et al. [42] proposed deformable DETR, where the attention module focuses only on key sampling points around a reference. Liu et al. [43] introduced a novel query formulation using dynamic anchor boxes in DETR and provided deeper insights into the role of queries in the model. Li et al. [44] presented a new denoising training approach to accelerate DETR's training process and enhance understanding of the slow convergence issue in similar Transformer-based methods. Zhang et al. [45] proposed DINO, a state-of-the-art end-to-end detector that enhances DETR with denoising training through contrastive methods, the hybrid query selection for anchor initialization, and a dual-forward scheme for box prediction. The models based on the Transformer architecture rely heavily on large-scale training data. Consequently, the intelligent interpretation in the remote sensing domain has driven deeper research into hyperspectral super-resolution. Li et al. [46] enhanced the deep image prior to the hyperspectral super-resolution which can obtain high-quality reconstruction results. Li et al. [47], [48] designed a spectral super-resolution framework by learning cross-scale relationships, achieving satisfactory results. More recently, they have integrated spectral unmixing into the super-resolution process, proposing an effective coupled unmixing framework [49].

B. Open-Vocabulary Object Detection

Recently, with the rapid advancement of vision–language pretraining, the open-vocabulary setting has emerged. These

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

new approaches aim to detect and recognize categories beyond the annotated label space.

Du et al. [50] introduced a novel method called detection prompt (DetPro), which learns continuous prompt representations for open-vocabulary object detection using a pretrained vision–language model. Feng et al. [51] proposed PromptDet, a two-stage open-vocabulary object detector that addresses the gap between image and region classification by adding learnable prompts when encoding class names, namely, regional prompt learning (RPL), which is designed to generalize from base-to-novel categories. Bravo et al. [52] proposed LocOv, a two-stage method for improving the matching between image and text representations: localized semantic matching (LSM) learns object semantics by matching image regions to words in captions, and specialized task tuning (STT) refines visual features for object detection using known object annotations. Wu et al. [53] proposed CORA, a DETR-style framework that adapts CLIP for open-vocabulary detection by using region prompting to address the whole-to-region distribution gap and anchor prematching to enhance the object localization through a class-aware matching mechanism. Gao et al. [54] proposed a method to automatically generate pseudobounding-box annotations of diverse objects from large-scale image–caption pairs by leveraging the localization ability of pretrained vision–language models, which are then directly used for training object detectors. Kim et al. [55] presented RO-ViT, a contrastive image–text pretraining approach to bridge the gap between image-level pretraining and open-vocabulary object detection.

C. Visual-Language Model

Recent advancements in vision–language models highlight that enhanced pretraining strategies can significantly improve the semantic understanding of visual inputs, bridging the gap between visual and textual information.

Radford et al. [56] introduced CLIP, an efficient and scalable method for natural language supervision that combines textual and visual data through contrastive learning for language-image pretraining. Jia et al. [57] proposed align, which leverages a noisy dataset of over one billion image–text pairs without data filtering or postprocessing, using contrastive learning loss and a simple dual-encoder structure to align visual and textual representations. Li et al. [58] introduced fast language-image pretraining (FLIP), a more efficient method for training CLIP that involves randomly masking and removing a large portion of image patches to learn from more image–text pairs in the same wall-clock time and to contrast more samples per iteration with a similar memory footprint. Mu et al. [59] investigated the potential of self-supervised learning to enhance the language supervision for visual representation learning and introduce SLIP, a multitask learning framework that combines self-supervised learning with CLIP pretraining. Yu et al. [60] introduced contrastive captioner (CoCa), a streamlined image–text encoder–decoder model that leverages both contrastive and captioning losses to integrate the strengths of contrastive and generative methods. Li et al. [61] introduced BLIP-2, a universal and efficient pretraining strategy that utilizes off-the-shelf frozen image encoders and large language models for vision–language pretraining, bridging the modality gap with a lightweight querying Transformer.

III. Methodology

A. Overview

The LLaMA-Unidetector framework is designed to tackle the challenges of open-vocabulary object detection by decoupling object localization and category recognition into two distinct stages. In the object localization stage, the class-agnostic detector distinguishes between foreground and background without differentiating specific predefined classes, allowing the model to localize targets from categories that are not predefined. In the category recognition stage, an MLLM leverages the powerful reasoning ability of the text modality to perform classification on the foreground regions. This decoupled learning strategy enables the independent optimization of each stage, leveraging task-specific datasets and methodologies to maximize performance and scalability. The pseudo-code is as shown in Algorithm 1.

In the first stage, the class-agnostic detector distinguishes between foreground and background without differentiating specific classes, allowing the model to localize targets from categories that are not predefined, thereby promoting robust generalization across different scenes and unseen object categories. The second stage employs TerraOV-LLM, an MLLM, to classify the detected objects. By combining visual features extracted from a pretrained vision encoder with textual prompts, TerraOV-LLM leverages its powerful reasoning and generalization capabilities of the text modality to recognize both base and novel categories. Compared with coupled models, this modular design jointly enhances the model's generalization ability while also offering flexibility and adaptability, making it well-suited for diverse scenarios, including zero-shot and open-world detection.

The inference pipeline of LLaMA-Unidetector, as illustrated in Fig. 3, consists of a class-agnostic detector and a large language model, TerraOV-LLM, which is collectively constituted by all subsequent modules. While the class-agnostic detector strengthens the model's ability to generalize, TerraOV-LLM supports zero-shot transfer, allowing recognition to extend from base-to-novel classes, previously unseen categories.

Sections III-B–III-D delve into the details of each stage and the overall learning strategy, providing a comprehensive understanding of the proposed framework.

B. Class-Agnostic Detection

To address the challenge of detecting unseen objects in open-world scenarios, we propose a class-agnostic detection strategy as the first stage of the LLaMA-Unidetector framework. This stage focuses solely on object localization, distinguishing between foreground and background without further class-specific differentiation. By decoupling localization from THe recognition, we establish a robust foundation for the subsequent recognition phase.

We adapt a standard object detector with two key branches, as depicted in Fig. 4: a localization branch for bounding box regression and a classification branch for object classification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

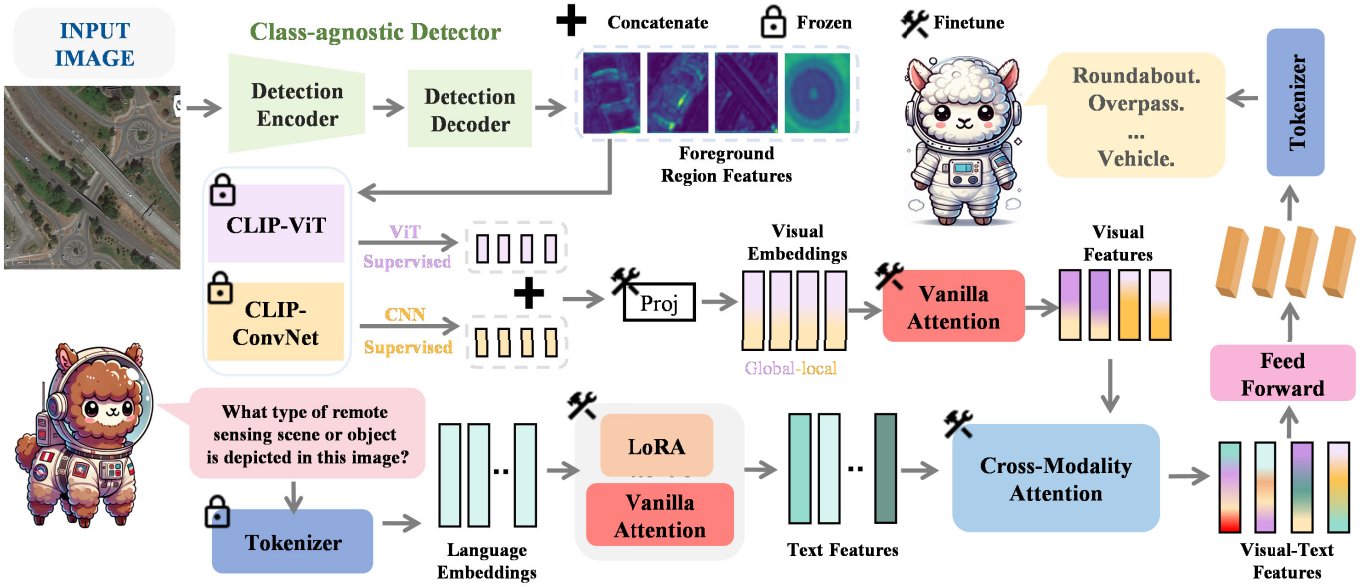


Fig. 3. Inference pipeline of LLaMA-Unidetector consists of a class-agnostic detector and a large language model, TerraOV-LLM, which is collectively constituted by all subsequent modules. As an open-vocabulary object detector, LLaMA-Unidetector adopts text as input and performs cross-modal fusion between image and text features. While the class-agnostic detector enhances the model's generalization capabilities, TerraOV-LLM enables zero-shot transfer, facilitating recognition from base classes to previously unseen novel categories.

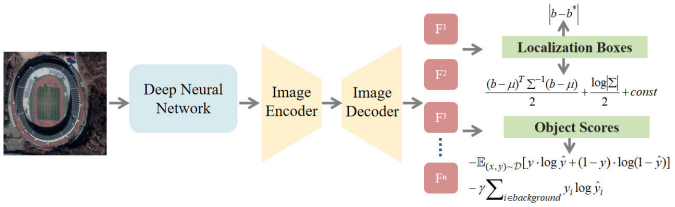


Fig. 4. Class-agnostic detector: focuses on foreground–background detection without performing fine-grained category differentiation.

The foreground–background classification is modeled using a probabilistic approach, where the predicted probability $\hat{y}$ of being foreground is given by the following equation:

$$\hat{y} = \sigma(\mathbf{W}_c \mathbf{f}_i + b) \tag{6}$$

where $\mathbf{f}_i$ represents the extracted object features, σ denotes the Sigmoid function, $\mathbf{W}_c$ is the foreground–background classification weights, and b represents the bias term. To improve robustness, an entropy regularization term is introduced to penalize uncertain predictions for background regions. The overall classification loss can be expressed as follows:

$$\mathcal{L}_{\text{FBclassification}} = -\mathbb{E}_{(\mathbf{x},y)\sim\mathcal{D}}\left[y \cdot \log \hat{y} + (1-y)\cdot \log\left(1-\hat{y}\right)\right] + \gamma \mathcal{H}_{\text{background}} \tag{7}$$

where $\mathcal{H}_{\text{background}} = -\sum_{i\in\text{background}} y_i \log \hat{y}_i$ is the entropy term and γ controls the regularization strength. By minimizing background entropy, the background becomes more certain, reducing the influence of noise on the foreground and enhancing the distinction between foreground and background.

For bounding box regression, the localization branch minimizes the geometric discrepancy between predicted and ground-truth boxes. We use the generalized intersection over union (GIoU) as the primary metric, defined as follows:

$$\mathcal{L}^{\text{GIoU}}_{\text{localization}} = 1 - \text{GIoU}\left(\mathbf{b}, \mathbf{b}^*\right) \tag{8}$$

where GIoU captures both the overlap and the shape alignment between the predicted box $\mathbf{b}$ and the ground-truth box $\mathbf{b}^*$.

To further incorporate uncertainty, bounding boxes are additionally modeled as Gaussian distributions $\mathcal{N}(\mathbf{b}|\mu, \Sigma)$, where μ is the predicted box center and Σ is the covariance matrix capturing prediction uncertainty. The GIoU localization loss then takes the form

$$\mathcal{L}^{\text{GIoU}}_{\text{localization}} = \frac{1}{2}(\mathbf{b}-\mu)^\top \Sigma^{-1}(\mathbf{b}-\mu) + \frac{1}{2}\log|\Sigma| + \text{const}. \tag{9}$$

In addition, we employ $L1$ loss to constrain the absolute coordinate differences between the predicted and ground-truth bounding boxes, thereby enhancing the localization accuracy of the detected objects

$$\mathcal{L}^{L1}_{\text{localization}} = \left|\mathbf{b} - \mathbf{b}^*\right|. \tag{10}$$

In our framework, we modify the classification branch to predict only a binary label—foreground or background—rather than classifying objects into specific categories. This simplification enables the model to generalize beyond base categories and detect novel objects effectively.

Formally, given an input image $\mathbf{I}$, the localization branch predicts a set of bounding boxes $\{\mathbf{b}_i\}$, while the modified classification branch outputs a binary label $c_i \in \{0, 1\}$ for each box, where $c_i = 1$ indicates foreground and $c_i = 0$ indicates background. The final detection results are represented as follows:

$$\mathcal{D} = \{(\mathbf{b}_i, c_i) | c_i = 1\}. \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

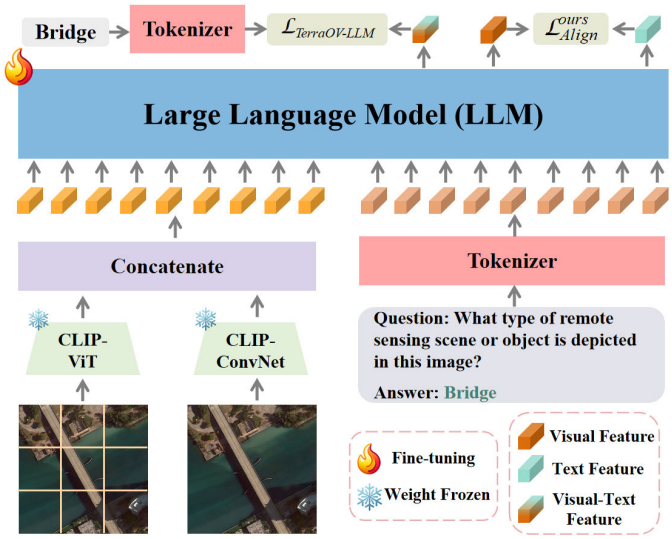


Fig. 5. Architecture of TerraOV-LLM, an MLLM enabling robust zero-shot and novel-class object recognition.

The overall training loss $\mathcal{L}$ consists of two components: the localization loss to optimize the bounding box predictions, and the classification loss to distinguish foreground from background

$$\mathcal{L} = \mathcal{L}^{L1}_{\text{localization}} + \mathcal{L}^{\text{GIoU}}_{\text{localization}} + \lambda \mathcal{L}_{\text{FBclassification}} \tag{12}$$

where λ is a balancing hyperparameter. The localization loss is implemented as $L1$ and GIoU loss, while the classification loss uses binary cross-entropy (BCE) with an entropy regularization term to optimize the foreground–background discrimination task.

This design removes the need for multiclass predictions, reducing model complexity and focusing the classification branch on the essential task of foreground–background discrimination.

1) Advantages: By training the model to distinguish only between foreground and background, the classification branch avoids overfitting to base categories, improving its generalization to unseen classes. This approach mitigates the label bias inherent in traditional object detectors trained on predefined categories, such as those in COCO [10], and makes the detector more versatile. In addition, the binary classification task simplifies training, allowing for faster convergence and reduced computational overhead.

2) Decoupling Benefits: The decoupling localization and recognition offers several key benefits. First, it allows for independent optimization of localization without being constrained by recognition, ensuring high-quality bounding box predictions. Second, the detected foreground objects can be processed by a separate recognition model, such as TerraOV-LLM, which is tailored for category generalization. Finally, this modular design facilitates the integration of advanced recognition methods, supporting scalability and adaptability for diverse downstream tasks.

C. Multimodal Recognition With TerraOV-LLM

In the second stage of LLaMA-Unidetector, we perform category recognition using TerraOV-LLM, as shown in Fig. 5, an MLLM that bridges vision and language modalities. This stage leverages the strong generalization capabilities of LLMs to classify foreground objects detected in the first stage, even in zero-shot and novel-class scenarios. By integrating pretrained vision encoders with LLMs, we enable the model to infer semantic information about unseen objects, establishing a robust framework for open-world object recognition.

1) Input Construction and Multimodal Fusion: To construct inputs for TerraOV-LLM, we first extract visual features from the detected bounding box tensors produced by Stage 1.

1) *Visual Feature Extraction:* For each bounding box $\mathbf{b}_i$ detected in Stage 1, the corresponding feature tensor $\mathbf{f}_i$ is extracted directly from the feature map of the input image $\mathbf{I}$ processed in the first stage. These feature tensors retain high-level semantic information and spatial context without requiring additional cropping or resizing operations. We use a pretrained CLIP encoder to project $\mathbf{f}_i$ into a compact visual feature vector

$$\mathbf{v}_i = \text{CLIP}(\mathbf{f}_i) \tag{13}$$

where $\mathbf{v}_i \in \mathbb{R}^d$ is the visual feature vector of the detected object.

2) *Prompt Design for Language Inputs:* To incorporate the language understanding, we design task-specific textual prompts such as "what category does the object in this bounding box belong to?." These prompts are tokenized using the LLM tokenizer, producing token embeddings $\mathbf{t} \in \mathbb{R}^{l \times e}$, where l is the prompt length and e is the embedding dimension.

3) *Visual–Textual Feature Alignment:* The cross-modal contrastive learning aligns the similarity between visual and textual features to enhance the performance of multimodal models. Specifically, it aims to maximize the similarity between related modality pairs (e.g., an image and its textual information) while minimizing the similarity between unrelated modality pairs. This alignment occurs in the prestep of cross-modal attention, where visual features are aligned with semantic textual features to ensure that they share similar representations in the feature space, as expressed by the following formula:

$$\mathcal{L}^{\text{ours}}_{\text{align}} = \mathbb{E}_{(\mathbf{v},\mathbf{t})\sim\mathcal{D}}\left[-\log \frac{\exp(\text{sim}(\mathbf{v},\mathbf{t})/\tau)}{\sum_{t'\in\mathcal{T}} \exp(\text{sim}(\mathbf{v},t')/\tau)}\right] \tag{14}$$

where $\text{sim}(\cdot,\cdot)$ denotes the cosine similarity, τ is a temperature scaling factor, and $\mathcal{T}$ is the set of textual prompts. $\mathbf{v}$ represents the feature of the object category, $\mathbf{t}$ denotes the corresponding textual feature of that category, and $\mathbf{t}'$ refers to the textual information of any category in the set $\mathcal{T}$. It facilitates the subsequent visual–textual multimodal fusion. Visual and textual modalities carry rich yet distinct information. The visual information captures low-level details such as color, shape, and texture, as well as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

high-level objects and scenes, while textual information provides abstract and semantic descriptions. The cross-modal contrastive learning ensures semantic consistency between the visual and textual modalities, thereby improving zero-shot learning and transfer capabilities, allowing the model to understand and recognize novel object categories. Furthermore, it enhances performance across multimodal tasks such as image–text matching, facilitates more efficient utilization of semantic information, and allows the strengths of both modalities to complement each other. Ultimately, this approach improves robustness and generalization, enabling the model to adapt to diverse tasks and noisy data.

4) *Multimodal Fusion:* The visual features $\mathbf{v}_i$ and token embeddings $\mathbf{t}$ are linearly projected into a shared latent space and concatenated to form a multimodal tensor

$$\mathbf{z}_i = [\mathbf{W}_v \mathbf{v}_i; \mathbf{W}_t \mathbf{t}] \tag{15}$$

where $\mathbf{W}_v$ and $\mathbf{W}_t$ are the learnable projection matrices. The combined representation $\mathbf{z}_i$ is then processed by the LLM's transformer layers for joint reasoning.

2) Recognition With TerraOV-LLM: Once the multimodal features are fused, TerraOV-LLM performs end-to-end reasoning to infer the category of the object. The LLM processes the combined representation $\mathbf{z}_i$ and directly outputs the predicted category $\hat{y}_i$ in natural language form. This output is mapped back to the predefined category set or interpreted as an open-world prediction. Formally,

$$\hat{y}_i = \mathrm{LLM}(\mathbf{z}_i) \tag{16}$$

where $\hat{y}_i$ is a textual response (e.g., *"plane"* or *"vehicle"*), which can optionally be matched to predefined labels for evaluation. The reasoning process leverages the semantic knowledge encoded in the LLM, enabling it to recognize both base and novel categories.

3) Advantages of TerraOV-LLM: The integration of TerraOV-LLM into the LLaMA-Unidetector framework provides several key benefits are listed as follows.

1) *Strong Generalization:* Leveraging the semantic understanding of LLMs enables robust performance in zero-shot and novel-class detection tasks.
2) *Vision–Language Alignment:* The multimodal fusion mechanism aligns visual and textual modalities, incorporating the contextual information from prompts for improved reasoning.
3) *Prompt Flexibility:* The natural language prompts allow the system to adapt to diverse tasks without retraining, offering scalability and versatility.
4) *Pretrained Knowledge Utilization:* The CLIP encoder provides robust visual feature extraction, while the LLaMA backbone contributes extensive semantic knowledge, resulting in a synergistic multimodal model.
5) *Domain Adaptation:* Fine-tuning on remote sensing datasets ensures that TerraOV-LLM is tailored to the unique characteristics of aerial and satellite imagery.

Algorithm 1 Pseudocode of LLaMA-Unidetector

Input: Training epoch T, Detection Batchsize N_d, Multimodal LLM Batchsize N_m, Class-agnostic Data $\mathcal{D}$ for Large-scale Detection Pretraining, VQA Data $\mathcal{VQA}$ for Large-scale Multimodal LLM Pretraining, Class-agnostic Detection Encoder e, Class-agnostic Detection Decoder d, MLLM Image Processing Branch and Text Processing Branch $TerraOV_{IP}$, $TerraOV_{TP}$, Cross Modality Attention CMA, Feed Forward Network FFN

for $t = 1, 2, 3, \ldots, T$ **do**
  for *randomly sampled minibatches* $\{x_{di}\}_{i=1}^{N_d} \in \mathcal{D}$ **do**
    for *ALL* $i \in \{1, 2, \ldots, N_d\}$ **do**
      $e_{di} = e(x_{di})$
      $d_{di} = d(x_{li})$
    end
    Get boxes and foreground confidence scores from the class-agnostic detection;
    Compute loss: $\mathcal{L}_{\text{Localization}}^{\text{GIoU}}$;
    Compute loss: $\mathcal{L}_{\text{Localization}}^{\text{L1}}$;
    Compute loss: $\mathcal{L}_{\text{FBclassification}}$;
    Update the network;
  end
end
for $t = 1, 2, 3, \ldots, T$ **do**
  for *randomly sampled minibatches* $\{x_{mi} \in\}_{i=1}^{N_m} \in \mathcal{VQA}$ **do**
    for *ALL* $i \in \{1, 2, \ldots, N_m\}$ **do**
      $zv_{mi} = TerraLLM_{IP}(x_{mi})$;
      $zt_{mi} = TerraLLM_{TP}(x_{mi})$;
      $z_{mi} = FFN(CMA(zv_{mi}, zt_{mi}))$;
    end
    Get answering from the Multimodal large language model;
    Compute loss: $\mathcal{L}_{\text{align}}^{\text{ours}}$;
    Compute loss: $\mathcal{L}_{\text{TerraOV-LLM}}$;
    Update the network;
  end
end

Consequently, the multimodal large model loss is defined as follows:

$$\mathcal{L}_{\text{TerraOV-LLM}} = -\sum_{i=1}^{C} \mathrm{TKN}(y_i) \log(\sigma(\mathrm{FFN}(\mathrm{CMA}(v, t)))). \tag{17}$$

In this context, $\mathrm{TKN}(y_i)$ denotes the tokenized ground-truth labels, which represent the textual descriptions of the target classes. $\mathrm{CMA}(v, t)$ refers to the cross-modality attention module, which is utilized to fuse the visual features v and the textual features t. FFN stands for the feed-forward network, which further processes the fused features. σ represents the Sigmoid function, mapping the output of the feed-forward

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

network to the range [0, 1], indicating the predicted class probabilities. C denotes the number of classes.

The integration of TerraOV-LLM transforms object recognition into a multimodal reasoning problem, leveraging the complementary strengths of vision and language models. By incorporating pretrained encoders, prompt-based learning, and domain-specific fine-tuning, TerraOV-LLM bridges the gap between open-vocabulary object detection and real-world applicability in remote sensing.

D. Learning Strategy and Optimization

To ensure robust performance in both localization and recognition tasks, we adopt a decoupled learning strategy, where the localization and recognition branches are trained separately, each optimized for its specific goal. This strategy not only simplifies the training process but also allows for leveraging diverse datasets tailored to each stage, maximizing the potential of the framework in open-world scenarios. The overall optimization objective of LLaMA-Unidetector unifies these components

$$L_{\text{ours}} = \begin{cases} L_{\text{localization}}^{\text{GIoU}} + L_{\text{localization}}^{L^1} + \lambda L_{\text{FB classification}}, & \text{stage}_1 \\ L_{\text{align}}^{\text{ours}} + L_{\text{TerraOV-LLM}}, & \text{stage}_2 \end{cases} \tag{18}$$

where λ is a balancing hyperparameter.

1) Stagewise Training: The decoupled learning strategy consists of the following two stages.

1) *Localization Branch Pretraining:* The localization branch is trained on a large-scale object detection dataset, where the labels are adjusted to a class-agnostic format (foreground versus background). To ensure adaptability to remote sensing data, additional class-agnostic annotations from remote sensing datasets are included during pretraining. For domain-specific tasks, further fine-tuning on class-agnostic versions of task-specific datasets can be performed, but this step is optional for zero-shot or open-world scenarios.
2) *Recognition Branch Pretraining:* The recognition branch is designed as a multimodal reasoning model, trained in a visual question answering (VQA) format on extensive classification datasets. During training, the detected object regions are paired with textual prompts (e.g., "what category does this object belong to?"), enabling the MLLM to generalize across diverse classes and domains.

2) Inference Pipeline: During inference, the two branches are seamlessly integrated. The localization branch outputs bounding box predictions and their corresponding feature tensors, which are then passed to the recognition branch for category prediction, as described in Section III-A.

3) Ensuring Consistency Between Stages: To maintain consistency between the localization and recognition stages, we propose the following strategies.

1) *Feature Alignment:* During training, ensure that the localization branch produces feature tensors that are compatible with the input requirements of the recognition branch. This can be achieved by aligning the output feature space of the localization branch with the input space of the recognition branch using a learnable projection matrix

$$\mathbf{f}_i' = \mathbf{W}_p \cdot \mathbf{f}_i \tag{19}$$

where $\mathbf{f}_i'$ is the aligned feature, $\mathbf{f}_i$ is the original feature tensor from the localization branch, and $\mathbf{W}_p$ is a learnable projection matrix.
2) *Unified Data Augmentation:* Employ shared data augmentation techniques (e.g., random crops, flips, and rotations) across both stages to ensure consistent data distributions, minimizing discrepancies between training and inference.

4) Advantages of Decoupled Learning: The decoupled learning strategy offers several advantages that are listed as follows.

1) *Specialized Training:* By independently optimizing the localization and recognition branches, each can be tailored to its specific task, leveraging datasets and techniques best suited to its requirements.
2) *Scalability:* The modular design allows for easy adaptation to new datasets or tasks by fine-tuning only the relevant branch without retraining the entire framework.
3) *Open-World Generalization:* The recognition branch, trained on diverse VQA tasks, is inherently capable of handling novel categories, while the localization branch ensures high-quality bounding box predictions across domains.

IV. Experiments

A. Datasets

To validate the effectiveness of our proposed LLaMA-Unidetector, we conducted extensive experiments on several publicly available remote sensing object detection datasets.

1) NWPU-VHR10: The NWPU-VHR10 [33] dataset is a widely used benchmark for object detection in high-resolution remote sensing images. It consists of 800 images, each with a resolution of 1000 × 1000 pixels, acquired from various sensors. The dataset includes 3651 annotated instances across ten object categories: airplane (APL), ship (SH), storage tank (STO), baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GTF), harbor (HA), bridge (BR), and vehicle (VE). The objects in the dataset vary significantly in scale, orientation, and appearance, presenting challenges for detection models. The images are captured under diverse conditions, further contributing to the complexity of the dataset. NWPU-VHR10 is commonly used for evaluating object detection algorithms in high-resolution remote sensing imagery.

2) DOTA1.0: DOTA-v1.0 [34] is a comprehensive object detection dataset for optical remote sensing images, featuring 2806 images with varying dimensions ranging from 800 to 4000 pixels in both width and height. The dataset encompasses a total of 188 282 annotated instances across 15 categories: plane (PL), BD, bridge (BR), GTF, small vehicle (SV), large

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

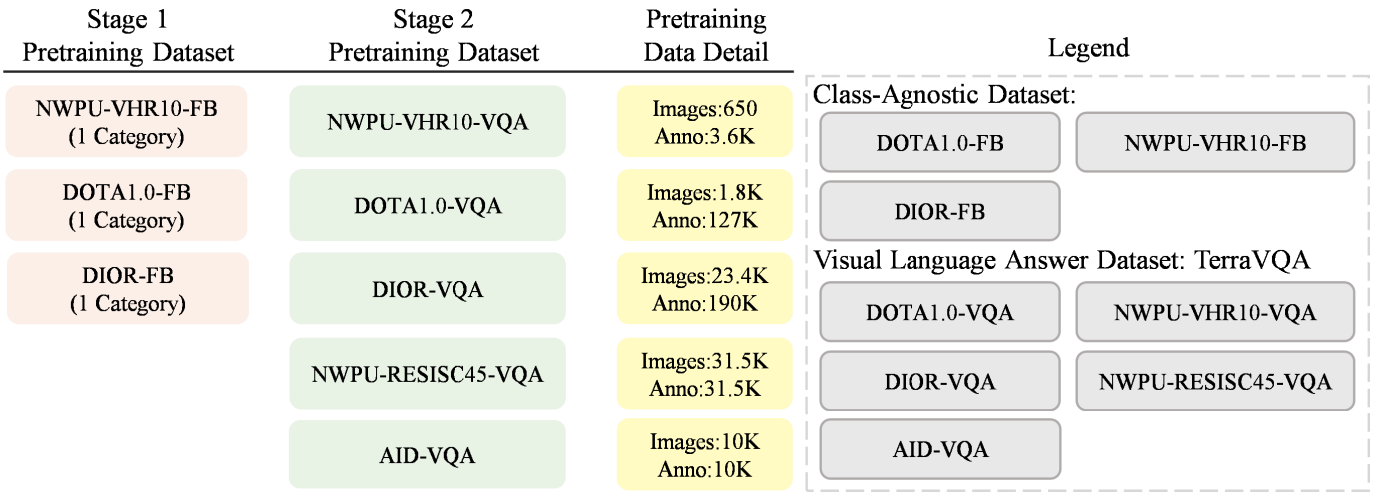


Fig. 6. Datasets employed in stage 1 and stage 2 of the pretraining framework for the class-agnostic detector and MLLM (TerraOV-LLM), encompassing foreground–background detection and VQA datasets with comprehensive details and legends.

vehicle (LV), ship (SH), TC, BC, STO, soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). It is divided into training, validation, and testing sets, containing 1411, 458, and 937 images, respectively.

3) DIOR: The DIOR [8] dataset is a large-scale benchmark for object detection in remote sensing imagery, collected from Google Earth. It contains 23 463 images and 192 472 annotated instances across 20 object categories, including airplane (APL), airport (APT), baseball field (BF), BC, bridge (BR), chimney (CH), dam (DAM), expressway service area (ESA), expressway toll station (ETS), harbor (HA), golf course (GF), GTF, overpass (OP), ship (SH), stadium (STA), STO, TC, train station (TS), vehicle (VE), and windmill (WM). The images have a resolution of 800 × 800 pixels, with spatial resolutions ranging from 0.5 to 30 m, and exhibit significant variation in object scales. The dataset is divided into 5862 training images, 5863 validation images, and 11 738 testing images. In this study, we fine-tune our model using a combination of the training and validation sets and evaluate it on the testing set.

B. Experimental Setting and Pretraining Data

1) Experimental Setting: Object detection models perform well when localizing and classifying objects seen during training. However, due to the difficulty and cost of creating and annotating detection datasets, trained models are typically limited to detecting a finite number of object categories. Unseen objects are often treated as background, which limits the applicability of traditional detectors in real-world remote sensing scenarios.

Open-vocabulary methods leverage MLLMs to utilize image–text pairs for pretraining, benefiting from the rich vocabulary provided by textual descriptions. During training, the model aligns these rich textual descriptions with corresponding images, thereby learning an effective visual-semantic space that establishes associations between base classes and novel classes. To evaluate the detector's capability to recognize unseen objects, we train the model on known base classes and test it on a set containing partially unseen novel classes. The category vocabulary for these novel classes partially overlaps between the training and testing datasets.

2) Pretraining Data: The pretraining of the class-agnostic detector primarily relies on detection datasets such as NWPU-VHR10-FB, DOTA1.0-FB, and DIOR-FB, as specified in pretraining stage 1 of Fig. 6. In addition, we convert remote sensing datasets, including NWPU-VHR10 [33], DOTA1.0 [34], DIOR [8], NWPU-RESISC45 [68], and AID [69], into a VQA format. For example, the data are structured as follows: *id: 180515, images: 03321_3_tenniscourt.jpg, "conversations": ["from": "human" and "value": "*`<image>\n`*Based on the visual features in the image, which category or object do you think this scene belongs to?," "from": "gpt" and "value": "tenniscourt"].* This results in the construction of TerraVQA, as illustrated in Fig. 7, a large-scale remote sensing VQA dataset, which broadly encompasses the detection VQA (DVQA) datasets, such as NWPU-VHR10-VQA, DOTA1.0-VQA, and DIOR-VQA, as well as the recognition VQA (RVQA) datasets, namely, NWPU-RESISC45-VQA and AID-VQA.

C. Implementation Details and Evaluation Metrics

1) Implementation Details: We implement our class-agnostic detection with Pytorch on four Tesla A6000 GPUs for 50 epochs. The input images are transformed to 800 × 800. The stochastic gradient descent (SGD) [70] with momentum 0.9 and weight decay 1×10^{-4} is selected as the optimizer. In addition, ResNet-50 [62] is adopted as the backbone, and the batch size is set to 9 per GPU, with an initial learning rate of 0.0001. The learning rate is decayed by a factor of 10 at the 11th, 33th, and 45th epoch, respectively.

In the MLLM recognition stage, we implement fine-grained recognition of foreground objects using PyTorch on four Tesla A6000 GPUs for fine-tuning over 5 epochs. We used the AdamW [71] optimizer with the following hyperparameters: learning rate $2e^{-5}$, weight decay 0.0005, betas (0.9, 0.95). The warmup epochs, batch size per GPU, and base learning rate are set to 2, 4, and 0.001, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

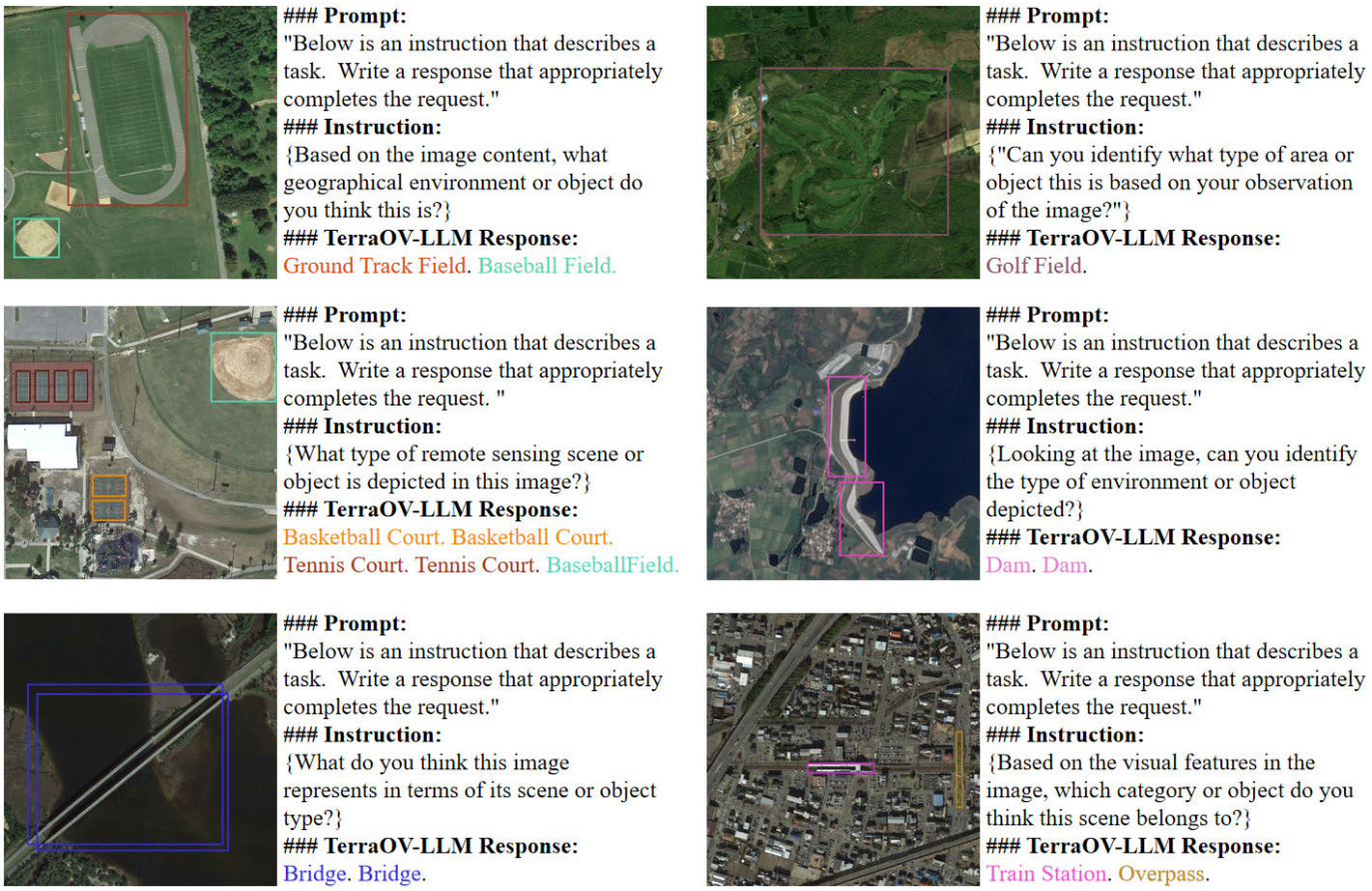


Fig. 7. Examples of LLaMA-Unidetector in action for open-vocabulary object detection, showcasing its powerful text comprehension and visual perception capabilities in remote sensing imagery.

2) Evaluation Metrics: For evaluating the performance of our model, we adopt the mean average precision (mAP) as the primary evaluation metric. Specifically, we report mAP at IoU = 0.50 (mAP50), where the average precision (AP) is computed for each class by considering a predicted bounding box a true positive if its intersection over union (IoU) with the ground-truth box exceeds 0.5. mAP50 refers to the mean of these AP values computed across all classes at this IoU threshold.

Mathematically, the AP for a class is calculated as follows:

$$\mathrm{AP} = \int_0^1 \mathrm{Precision}(r)dr \tag{20}$$

where precision(r) is the precision at a given recall level r. The precision is defined as follows:

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{21}$$

where TP refers to true positive and FP refers to false positive. The recall is defined as follows:

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{22}$$

where FN refers to false negative.

The mAP is then computed as the average of the AP values across all classes

$$\mathrm{mAP} = \frac{1}{C}\sum_{i=1}^{C} \mathrm{AP}_i \tag{23}$$

where C is the total number of classes. In this work, we report mAP at IoU = 0.50 (mAP50), where a predicted bounding box is considered a correct prediction if its IoU with the ground-truth box exceeds 0.5. AP50 refers to the AP computed at this IoU threshold.

D. Zero-Shot Transfer of LLaMA-Unidetector

We pretrain our model on large-scale datasets and directly evaluate it in a zero-shot setting on new datasets. To enable a more detailed comparison, we present several fine-tuned results to benchmark our model against prior works. Experimental results demonstrate the effectiveness of our algorithm in zero-shot scenarios and highlight its strong generalization ability for predicting novel classes.

1) NWPU-VHR10 Benchmark: The NWPU-VHR10 dataset contains fewer images and targets compared with the other two benchmarks, making it relatively easier. Since the categories in the DIOR and DOTA datasets encompass all the categories present in NWPU-VHR10, the challenge lies in the cross-domain setting, akin to zero-shot domain transfer. We configured the training and testing sets with a 1:1 ratio. We compared

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

TABLE I
MAP ON NWPU-VHR10 FOR OPEN-VOCABULARY OBJECT DETECTION

Model	Backbone	OD Pre-Training Data	Text Pre-Training Data	Zero Shot Transfer	Finetuning Val
Sparse R-CNN [39]	ResNet50 [62]	NWPU-VHR10 [33]	-	-	83.63
DINO [45]	ResNet50 [62]	NWPU-VHR10 [33]	-	-	84.40
YOLO-V10 [63]	CSPNet [64]	NWPU-VHR10 [33]	-	-	88.40
Grounding DINO [65]	SWIN-T [66]	DOTA1.0 [34],DIOR [8]	DOTA1.0 [34],DIOR [8]	72.96	80.62
YOLO-WORLD [67]	CSPNet [64]	DOTA1.0 [34],DIOR [8].	DOTA1.0 [34],DIOR [8].	74.40	89.19
LLaMA-Unidetector	ResNet50 [62]	DOTA1.0-FB,DIOR-FB	TerraVQA subset (excluding NWPU-VHR10-VQA)	75.46	89.52

TABLE II
MAP ON DOTA1.0 FOR OPEN-VOCABULARY OBJECT DETECTION

Model	Backbone	OD Pre-Training Data	Text Pre-Training Data	Zero Shot Val	Finetuning Val
Sparse R-CNN [39]	ResNet50 [62]	DOTA1.0 [34]	-	-	72.03
DINO [45]	ResNet50 [62]	DOTA1.0 [34]	-	-	76.50
YOLO-V10 [63]	CSPNet [64]	DOTA1.0 [34]	-	-	76.10
Grounding DINO [65]	SWIN-T [66]	NWPU-VHR10 [33],DIOR [8]	NWPU-VHR10 [33],DIOR [8]	30.21	74.58
YOLO-WORLD [67]	CSPNet [64]	NWPU-VHR10 [33] ,DIOR [8]	NWPU-VHR10 [33],DIOR [8]	36.04	73.66
LLaMA-Unidetector	ResNet50 [62]	NWPU-VHR10-FB DIOR-FB	TerraVQA subset (excluding DOTA1.0-VQA)	50.22	70.90

the performance of the proposed LLaMA-Unidetector with Grounding DINO and YOLO-WORLD on the NWPU-VHR10 dataset, as shown in Table I. We pretrained our class-agnostic detector using the DIOR-FB and DOTA1.0-FB datasets, and fine-tuned our TerraOV-LLM model using the DIOR-VQA, DOTA1.0-VQA, NWPU-RESISC45-VQA, and AID-VQA datasets. The model was directly tested on the NWPU-VHR10 benchmark. Under the zero-shot transfer setting, our LLaMA-Unidetector achieved an mAP of 75.46%, outperforming grounding DINO and YOLO-WORLD by 2.5% and 1.06%, respectively. Furthermore, we also fine-tuned our model on the NWPU-VHR10 dataset and trained three models based on CNN and Transformer architectures, including the closed-set classifiers Sparse-RCNN, DINO, and YOLOV10. Predefined-Class Object Detectors perform well on base classes but fail to recognize novel classes. After fine-tuning, both our LLaMA-Unidetector and YOLO-WORLD models demonstrated improvements in performance.

2) DOTA1.0 Benchmark: We pretrained our class-agnostic detector using the NWPU-VHR10-FB and DIOR-FB datasets and fine-tuned our TerraOV-LLM model using the NWPU-VHR10-VQA, DIOR-VQA, NWPU-RESISC45-VQA, and AID-VQA datasets. The model was rigorously evaluated on the prestigious DOTA1.0 benchmark, and the results, as detailed in Table II, demonstrate its superior performance. Under the zero-shot transfer setting, our LLaMA-Unidetector achieved an mAP of 50.22%, significantly outperforming Grounding DINO and YOLO-WORLD by 20.01% and 14.18%, respectively. In addition, we also fine-tuned the open-vocabulary object detector on the DOTA1.0 dataset and trained three models based on CNN and Transformer architectures, including the closed-set classifiers Sparse-RCNN, DINO, and YOLOV10. The results show that predefined-class object detectors perform well on base classes but are unable to recognize novel classes. After fine-tuning, the performance of the open-vocabulary object detector improved significantly. For the DINO model in the predefined-class object detector, our LLaMA-Unidetector model, along with the grounding DINO model, integrates text modality, shifting from a discriminative model to a generative model. As a result, the evaluation criterion for generative models is less favorable, and the mAP dropped by 5.6% and 1.92%, respectively, compared with DINO.

3) DIOR Benchmark: We pretrained our class-agnostic detector using the NWPU-VHR10-FB and DOTA1.0-FB datasets and fine-tuned our TerraOV-LLM model using the NWPU-VHR10-VQA, DOTA1.0-VQA, NWPU-RESISC45-VQA, and AID-VQA datasets. The model was rigorously assessed on the DIOR benchmark, with the results presented in Table III showcasing its superior performance. Under the zero-shot transfer setting, our LLaMA-Unidetector achieved an mAP of 51.38%, significantly outperforming grounding DINO and YOLO-WORLD by 24.3% and 23.35%, respectively. We also fine-tuned the model on the DIOR dataset and trained three models based on CNN and Transformer architectures, including the closed-set classifiers Sparse-RCNN, DINO, and YOLOV10. Predefined-class object detectors perform well on base classes but fail to recognize novel classes. Compared with the DINO model, our LLaMA-Unidetector and grounding DINO models exhibited a decrease in mAP of 7.36%

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

TABLE III
MAP ON DIOR FOR OPEN-VOCABULARY OBJECT DETECTION

Model	Backbone	OD Pre-Training Data	Text Pre-Training Data	Zero Shot Val	Finetuning Val
Sparse R-CNN [39]	ResNet50 [62]	DIOR [8]	-	-	74.78
DINO [45]	ResNet50 [62]	DIOR [8]	-	-	80.50
YOLO-V10 [63]	CSPNet [64]	DIOR [8]	-	-	81.86
Grounding DINO [65]	SWIN-T [66]	NWPU-VHR10 [33],DOTA1.0 [34]	NWPU-VHR10 [33],DOTA1.0 [34]	27.08	78.64
YOLO-WORLD [67]	CSPNet [64]	NWPU-VHR10 [33],DOTA1.0 [34]	NWPU-VHR10 [33],DOTA1.0 [34]	28.03	84.41
LLaMA-Unidetector	ResNet50 [62]	NWPU-VHR10-FB DOTA1.0-FB	TerraVQA subset (excluding DIOR-VQA)	51.38	73.14

TABLE IV
AP ON DIOR FOR OPEN-VOCABULARY OBJECT DETECTION

Method	Base										Novel									
	APL	BC	BF	BR	GTF	HA	SH	STO	TC	VE	APT	CH	DAM	ETS	ESA	GF	OP	STA	TS	WM
Grounding-DINO	0.73	0.78	0.03	0.16	0.45	0.43	0.93	0.79	0.76	0.28	0	0.02	0.02	0	0	0	0.02	0	0	0
YOLO-WORLD	0.52	0.89	0.03	0.26	0.65	0.49	0.92	0.65	0.91	0.26	0	0	0	0	0	0	0.01	0	0	0
Ours	0.72	0.76	0.68	0.25	0.64	0.35	0.73	0.53	0.81	0.30	0.76	0.34	0.54	0.06	0.55	0.61	0.38	0.53	0.09	0.63

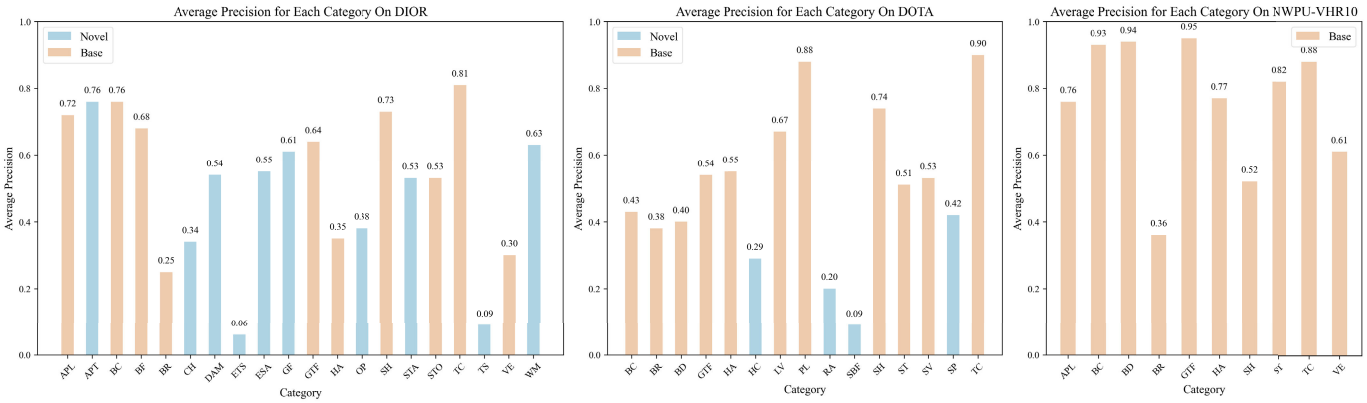


Fig. 8. AP for each category on the DIOR, DOTA, and NWPU-VHR10 datasets, demonstrating the model's performance on both base and novel categories.

and 1.86%, respectively. To better demonstrate the superiority of our LLaMA-Unidetector algorithm, we present the fine-grained AP values for our algorithm and two comparison methods on the DIOR dataset, as shown in Table IV. For the base classes, the performance of our algorithm is only slightly better than the other two methods, with negligible differences. However, for the novel classes, our algorithm exhibits a clear and overwhelming advantage over the other two methods.

4) AP of Fine-Grained Base and Novel Categories: To thoroughly assess the effectiveness of our open-vocabulary object detection model, LLaMA-Unidetector, we analyze its performance across various fine-grained categories in the DIOR, DOTA, and NWPU-VHR10 datasets, as illustrated in Fig. 8. On the DIOR dataset, the model performs exceptionally well on known categories, such as airplane, BC, and ship, with AP values exceeding 0.7, and achieving an impressive 0.81 on the TC category. However, the model's performance on unseen categories, such as ETS and TS, is notably poor, with AP values of only 0.06 and 0.09, respectively. For the DOTA dataset, the model achieves an AP of 0.88 on the known category plane and 0.9 on the TC category, but its performance on unseen categories, such as helicopter, roundabout, and soccer ball field, is suboptimal. In contrast, despite the limited sample size in the NWPU-VHR10 dataset, the model demonstrates relatively strong AP performance across its categories.

5) Recall Performance: For a more in-depth evaluation of the performance of our open-vocabulary object detection model, LLaMA-Unidetector, we conducted experiments to analyze its recall performance on both seen (base) and unseen (novel) categories across the DIOR and DOTA remote sensing datasets, as shown in Fig. 9. For seen categories, LLaMA-Unidetector achieves the highest recall rate, followed by YOLO-World and grounding DINO. For unseen categories, while our model experiences a slight decline in recall, YOLO-World and Grounding DINO exhibit a severe drop, with recall rates below 16%. In remote sensing scenarios, where the background is highly complex, the class-agnostic object detector exhibits its unique advantages.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

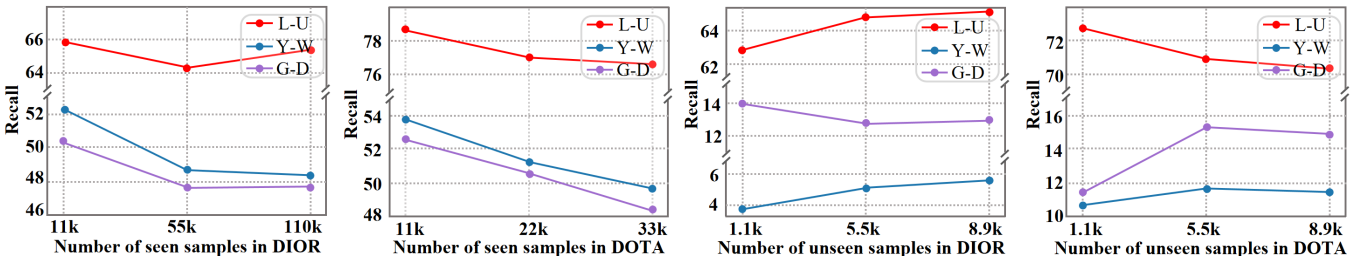


Fig. 9. Recall performance of the open-vocabulary object detection model on seen and unseen objects in the DOTA and DIOR datasets. L-M stands for LLaMA-Unidetector, Y-W denotes YOLO-WORLD, and G-D refers to Grounding-DINO.

By focusing solely on foreground–background distinction, the model can effectively locate a wide range of unseen objects, thereby enhancing its generalization ability, adaptability, and robustness, especially for unseen and novel categories. This makes it more effective in complex, large-scale remote sensing environments. In contrast, models like Grounding DINO and YOLO-World, which need to simultaneously optimize for both localization and recognition, are constrained during the localization process by the recognition task. This dual optimization may hinder the optimal regression of bounding boxes, limiting their performance. As the number of samples for seen categories increases, the model's recall rate shows a declining trend, with larger validation datasets more effectively reflecting the true performance of the model. For unseen categories, our model demonstrates a gradual increase in recall on the DIOR dataset, showcasing the superiority of the algorithm, whereas on the DOTA dataset, the recall exhibits a downward trend, which gradually flattens out, still achieving performance exceeding 70%.

E. Image-to-Text Generation With LLaMA-Unidetector Based on Instructional Prompts

LLaMA-Unidetector performs open-vocabulary object detection by leveraging both image and textual information. As illustrated in Fig. 7, given a detection image input, the model is provided with the prompt: "Below is an instruction that describes a task. Write a response that appropriately completes the request." Following this, an instruction such as "based on the image content, what geographical environment or object do you think this is?" is issued. To enhance the model's robustness, six distinct instructional prompts are used for fine-tuning, including examples such as: "looking at the image, can you identify the type of environment or object depicted?" and "what type of remote sensing scene or object is depicted in this image?." The MLLM, TerraOV-LLM, in conjunction with the class-agnostic detector's localization results, generates corresponding textual responses based on the instructions, inferring the relevant object information.

This approach demonstrates remarkable generalization capabilities, enabling the detection and identification of unseen and novel object classes. By combining image and text inputs, LLaMA-Unidetector effectively adapts to new, previously unseen categories without requiring additional training on those classes. Furthermore, the integration of TerraOV-LLM, a powerful MLLM, significantly enhances both text understanding and visual perception. TerraOV-LLM is not only able to process image data but also interpret textual instructions, generating accurate responses based on the visual context. This fusion of multimodal capabilities allows the model to perform complex reasoning and deliver precise object identification, making it highly robust and adaptable to diverse remote sensing imagery scenarios.

TABLE V
RECALL RATES OF CLASS-SPECIFIC AND CLASS-AGNOSTIC DETECTORS

Detector	Detection	GT	Recall
Class-Specific	115362	124440	92.70
Class-Agnostic	116049 (+687)		93.26 (+0.56)

F. Ablation Studies

In this section, we present ablation studies to systematically assess the contributions of individual components within the proposed LLaMA-Unidetector. The goal of this analysis is to gain a more comprehensive insight into the model's performance and to pinpoint specific areas for potential enhancement.

1) Superiority of Class-Agnostic Object Detector: To validate the effectiveness of the class-agnostic detection approach, we conducted an ablation study comparing the recall performance of class-specific and class-agnostic object detectors. "Class-specific" refers to a detector that can only detect predefined, known categories and is limited to using fixed-category datasets. The class-agnostic detector can enhance its ability to localize foreground targets by leveraging a large scale, specialized foreground–background dataset for training. This is a characteristic that class-specific detectors, which can only be trained on fixed-category datasets, lack—demonstrating the superior functionality of the class-agnostic detector. As shown in Table V, the DIOR dataset contains a total of 124 440 labeled instances. The class-specific detector detects 115 362 instances, achieving a recall rate of 92.70%. In comparison, the class-agnostic detector detects 116 049 instances, with a recall rate of 93.26%. This represents an increase of 687 additional detected targets, improving the recall rate by 0.56%.

2) Advantages of Decoupled Learning Architecture and Data Volume: MLLMs are data hungry, with their generalization ability improving as the amount of data increases. The VQA dataset generated from detection datasets is limited in size. To enhance the generalization capacity of the MLLM, we decoupled the model into two stages: the class-agnostic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

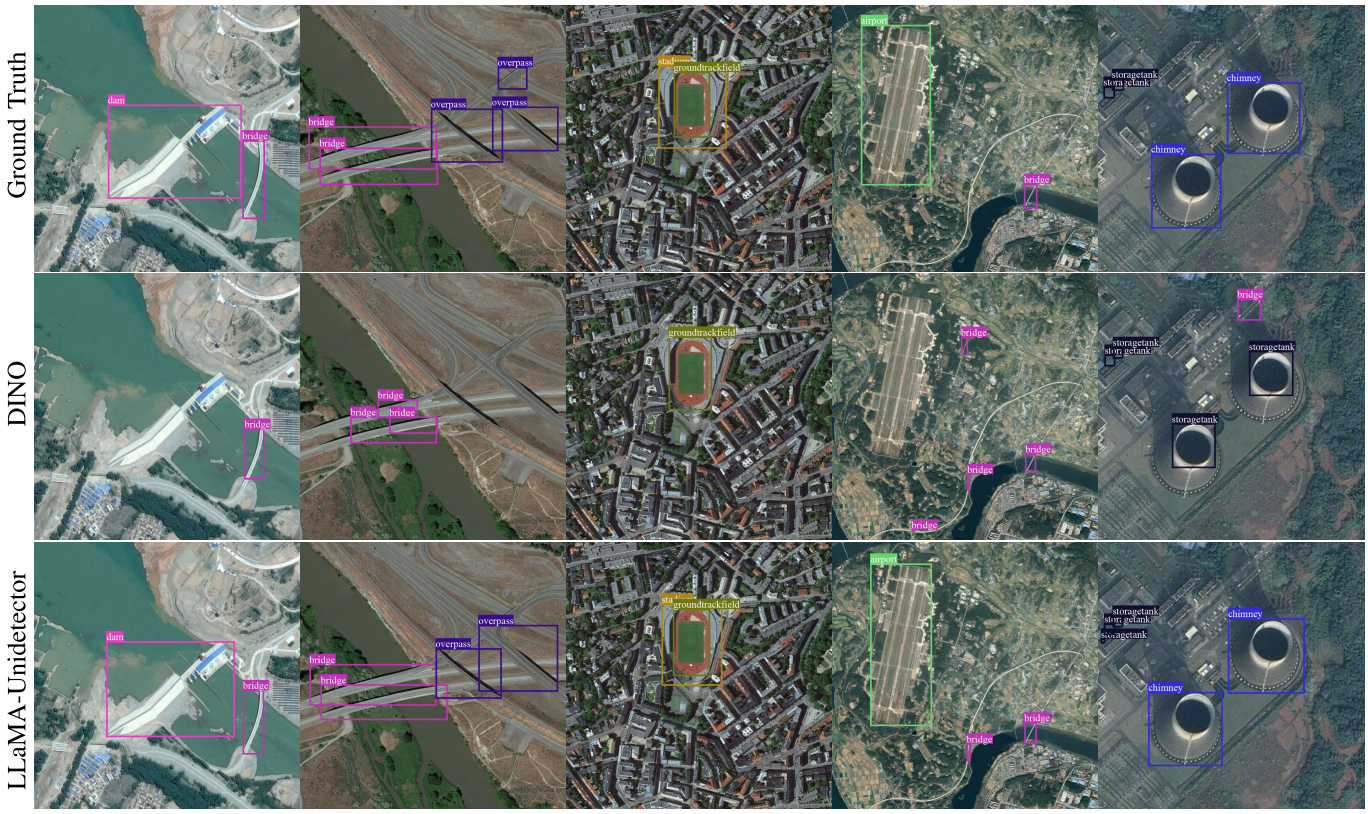


Fig. 10. Visualization of the performance of DINO and LLaMA-Unidetector models on DIOR under open-set conditions. Base categories: bridge, GTF, storage tank. Novel categories: dam, overpass, airport, and chimney. (a) Instance 1. (b) Instance 2. (c) Instance 3. (d) Instance 4. (e) Instance 5.

TABLE VI
MAP AFTER FINE-TUNING ON DVQA AND RVQA DATASETS

Dataset		mAP
DVQA	RVQA	
✓		50.58
✓	✓	51.38 (+0.8)

object detection stage and the multimodal language recognition stage, which allows for the incorporation of large-scale recognition datasets. In this context, we conducted an ablation study on the DIOR benchmark, as shown in Table VI. When the MLLM is trained using only the DVQA dataset, the mAP is 50.58%. However, when both DVQA and RVQA datasets are used, the mAP improves to 51.38%, demonstrating an enhancement of 0.8%. This indicates that as the data volume increases, the capability of the MLLM improves significantly, suggesting that an increase in the VQA data volume enhances the model's generalization ability.

3) Distinctive Strengths of Multimodal Large-Scale Language Model: To highlight the capability of the MLLM in recognizing unseen novel classes, we conducted ablation experiment comparing the traditional object detection model DINO and our open-vocabulary object detection model LLaMA-Unidetector on the DIOR benchmark.

We trained the traditional object detector DINO using the DOTA1.0 dataset and evaluated it on the DIOR dataset. Fig. 10(a) demonstrates that the traditional DINO detector successfully identifies the base category "bridge" but is unable to detect the novel category "airport." Conversely, our open-vocabulary detector, LLaMA-Unidetector, transcends the limitations of a predefined vocabulary, effectively identifying both "airport" and "bridge."

Fig. 10(e) highlights the visual similarity between "chimney" and "storage tank," both of which exhibit a circular appearance, thus presenting a challenge for differentiation. The traditional DINO detector can identify the predefined "storage tank" but fails to detect the novel category "chimney." In contrast, our open-vocabulary detector, LLaMA-Unidetector, not only surpasses the vocabulary constraints but also adeptly differentiates between "storage tank" and "chimney."

As illustrated in Fig. 10, it is apparent that DINO is proficient in detecting base categories such as "bridge," "GTF," and "storage tank" but is deficient in recognizing novel categories like "dam," "overpass," "airport," and "chimney," consequently misclassifying them as background. In stark contrast, our open-vocabulary detector, LLaMA-Unidetector, excels in detecting both base and novel categories, while also effectively distinguishing between visually ambiguous categories, representing a significant breakthrough in object detection.

4) Conclusive Insights: The ablation studies presented above provide compelling evidence for the advantages of our proposed LLaMA-Unidetector. First, the class-agnostic object detection framework demonstrates superior recall performance compared with the class-specific detector, highlighting the effectiveness of class-agnostic detection in identifying a broader range of objects. Second, the decoupled learning architecture, combined with the incorporation of large-scale

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

recognition data, significantly enhances the model's generalization ability, reinforcing the critical role of multimodal learning in improving performance. Finally, the MLLM's ability to recognize unseen novel classes underscores its unique strength over traditional object detection models. These findings collectively validate the effectiveness of our approach and provide valuable insights for advancing the open-vocabulary object detection task.

V. Conclusion

The open-vocabulary object detection enables the recognition and localization of an arbitrary number and variety of target objects without the need for manual expansion of annotated data, marking a significant shift from the traditional paradigm of remote sensing image object detection. Unlike conventional closed-set detection methods, open-vocabulary detection can identify and localize previously unseen categories, injecting new vitality and presenting unprecedented challenges to the field of remote sensing object detection. This approach not only addresses the issue of scarce annotated data but also provides greater flexibility and scalability for remote sensing image analysis, particularly in dynamic and diverse real-world environments. The contribution of this article lies in the proposal of a novel framework designed to enhance the robustness and generalization capabilities of models, facilitating the recognition of previously unseen fine-grained categories and pushing remote sensing image object detection toward higher levels of automation and intelligence. Through this innovation, we aim to offer valuable insights and new research directions for the future development of remote sensing image analysis.

References

[1] L. Yue, H. Shen, W. Yu, and L. Zhang, "Monitoring of historical glacier recession in yulong mountain by the integration of multisource remote sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 2, pp. 388–400, Feb. 2018.

[2] X. Meng, H. Shen, Q. Yuan, H. Li, L. Zhang, and W. Sun, "Pansharpening for cloud-contaminated very high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2840–2854, May 2019.

[3] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[4] Y. Yu, T. Gu, H. Guan, D. Li, and S. Jin, "Vehicle detection from high-resolution remote sensing imagery using convolutional capsule networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1894–1898, Dec. 2019.

[5] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[6] Z. Hu, D. Yang, K. Zhang, and Z. Chen, "Object tracking in satellite videos based on convolutional regression network with appearance and motion features," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 783–793, 2020.

[7] J. A. Peeling et al., "Applications of remote sensing for land use planning scenarios with suitability analysis," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 6366–6378, 2024.

[8] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[9] H. Cho, Y. Seo, B. V. K. V. Kumar, and R. R. Rajkumar, "A multi-sensor fusion system for moving object detection and tracking in urban driving environments," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2014, pp. 1836–1843.

[10] T. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, Zurich, Switzerland. Cham, Switzerland: Springer, 2014, pp. 740–755.

[11] S. Shao et al., "Objects365: A large-scale, high-quality dataset for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8430–8439.

[12] B. Xi et al., "Transductive few-shot learning with enhanced spectral–spatial embedding for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 34, pp. 854–868, 2025.

[13] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[14] S. Chen, L. Zhang, and L. Zhang, "MSDformer: Multiscale deformable transformer for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525614.

[15] H. Yu, Y. Tian, Q. Ye, and Y. Liu, "Spatial transform decoupling for oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 7, pp. 6782–6790.

[16] Y. Li, Q. Hou, Z. Zheng, M.-M. Cheng, J. Yang, and X. Li, "Large selective kernel network for remote sensing object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16794–16805.

[17] Y. Pu et al., "Adaptive rotated convolution for rotated object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 6589–6600.

[18] D. Wang et al., "MTP: Advancing remote sensing foundation model via multitask pretraining," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11632–11654, 2024.

[19] D. Wang et al., "Advancing plain vision transformer toward remote sensing foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5607315.

[20] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[21] X. Yang et al., "Learning high-precision bounding box for rotated object detection via Kullback–Leibler divergence," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2021, pp. 18381–18394.

[22] D. Liang et al., "Anchor retouching via model interaction for robust object detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619213.

[23] X. Gu, T.-Y. Lin, W. Kuo, and Y. Cui, "Open-vocabulary object detection via vision and language knowledge distillation," 2021, *arXiv:2104.13921*.

[24] L. Wang et al., "Object-aware distillation pyramid for open-vocabulary object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 11186–11196.

[25] Z. Mal et al., "Open-vocabulary one-stage detection with hierarchical visual-language knowledge distillation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14054–14063.

[26] H.-C. Cho, W. Young Jhoo, W. Kang, and B. Roh, "Open-vocabulary object detection using pseudo caption labels," 2023, *arXiv:2303.13040*.

[27] A. Zareian, K. D. Rosa, D. H. Hu, and S.-F. Chang, "Open-vocabulary object detection using captions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14393–14402.

[28] Y. Zhong et al., "RegionCLIP: Region-based language-image pretraining," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16793–16803.

[29] S. Wu, W. Zhang, S. Jin, W. Liu, and C. Change Loy, "Aligning bag of regions for open-vocabulary object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 15254–15264.

[30] Y. Zang, W. Li, K. Zhou, C. Huang, and C. C. Loy, "Open-vocabulary DETR with conditional matching," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Jan. 2022, pp. 106–122.

[31] W. Kuo, Y. Cui, X. Gu, A. Piergiovanni, and A. Angelova, "F-VLM: Open-vocabulary object detection upon frozen vision and language models," 2022, *arXiv:2209.15639*.

[32] C. Lin et al., "Learning object-language alignments for open-vocabulary object detection," 2022, *arXiv:2211.14843*.

[33] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[34] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.

[35] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2016.

[36] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[37] J. Redmon and A. Farhadi, "YOLO9000: Better, faster, stronger," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 7263–7271.

[38] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[39] P. Sun et al., "Sparse R-CNN: End-to-end object detection with learnable proposals," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14454–14463.

[40] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," 2019, *arXiv:1904.01355*.

[41] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[42] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[43] S. Liu et al., "DAB-DETR: Dynamic anchor boxes are better queries for DETR," 2022, *arXiv:2201.12329*.

[44] F. Li, H. Zhang, S. Liu, J. Guo, L. M. Ni, and L. Zhang, "DN-DETR: Accelerate DETR training by introducing query denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 13619–13627.

[45] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.

[46] J. Li, K. Zheng, L. Gao, L. Ni, M. Huang, and J. Chanussot, "Model-informed multistage unsupervised network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516117.

[47] J. Li, K. Zheng, W. Liu, Z. Li, H. Yu, and L. Ni, "Model-guided coarse-to-fine fusion network for unsupervised hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[48] J. Li, K. Zheng, J. Yao, L. Gao, and D. Hong, "Deep unsupervised blind hyperspectral and multispectral data fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[49] J. Li, K. Zheng, Z. Li, L. Gao, and X. Jia, "X-shaped interactive autoencoders with cross-modality mutual learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518317.

[50] Y. Du, F. Wei, Z. Zhang, M. Shi, Y. Gao, and G. Li, "Learning to prompt for open-vocabulary object detection with vision-language model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14084–14093.

[51] C. Feng et al., "PromptDet: Towards open-vocabulary detection using uncurated images," in *Proc. 17th Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 701–717.

[52] M. A. Bravo, S. Mittal, and T. Brox, "Localized vision-language matching for open-vocabulary object detection," in *Proc. DAGM German Conf. Pattern Recognit.* Cham, Switzerland: Springer, Jan. 2022, pp. 393–408.

[53] X. Wu, F. Zhu, R. Zhao, and H. Li, "CORA: Adapting CLIP for open-vocabulary detection with region prompting and anchor pre-matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7031–7040.

[54] M. Gao et al., "Open vocabulary object detection with pseudo bounding-box labels," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Jan. 2022, pp. 266–282.

[55] D. Kim, A. Angelova, and W. Kuo, "Region-aware pretraining for open-vocabulary object detection with vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 11144–11154.

[56] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[57] C. Jia et al., "Scaling up visual and vision-language representation learning with noisy text supervision," in *Proc. 38th Int. Conf. Mach. Learn.*, Jul. 2021, pp. 4904–4916.

[58] Y. Li, H. Fan, R. Hu, C. Feichtenhofer, and K. He, "Scaling language-image pre-training via masking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 23390–23400.

[59] N. Mu, A. Kirillov, D. Wagner, and S. Xie, "SLIP: Self-supervision meets language-image pre-training," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Oct. 2022, pp. 529–544.

[60] J. Yu, Z. Wang, V. Vasudevan, L. Yeung, M. Seyedhosseini, and Y. Wu, "CoCa: Contrastive captioners are image-text foundation models," 2022, *arXiv:2205.01917*.

[61] J. Li, D. Li, S. Savarese, and S. C. H. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn.*, Jan. 2023, pp. 19730–19742.

[62] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[63] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," 2024, *arXiv:2405.14458*.

[64] C.-Y. Wang, H.-Y. Mark Liao, Y.-H. Wu, P.-Y. Chen, J.-W. Hsieh, and I.-H. Yeh, "CSPNet: A new backbone that can enhance learning capability of CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 1571–1580.

[65] S. Liu et al., "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Nov. 2024, pp. 38–55.

[66] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[67] T. Cheng, L. Song, Y. Ge, W. Liu, X. Wang, and Y. Shan, "YOLO-world: Real-time open-vocabulary object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16901–16911.

[68] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[69] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[70] L. Bottou, "Stochastic gradient descent tricks," in *Neural Networks: Tricks Trade*, 2nd ed. Cham, Switzerland: Springer, 2012, pp. 421–436.

[71] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.



Guanqun Wang (Member, IEEE) was born in Heilongjiang, China, in 1995. He received the B.S. and Ph.D. degrees from Beijing Institute of Technology, Beijing, China, 2017 and 2023, respectively.

Currently, he was a Post-Doctoral with Peking University, Beijing. His research interests include large language model, object detection, and recognition.



Jianlin Xie received the B.S. degree in electronic information engineering from Beijing Institute of Technology, Beijing, China, in 2022, where he is currently pursing the Ph.D. degree in information and communication engineering.

His research interests include remote sensing object detection and recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.



Tong Zhang (Graduate Student Member, IEEE) was born in Hebei, China, in 1997. He received the Bachelor of Science degree from Hebei University of Technology, Tianjin, China, in 2019. He is currently pursuing the Ph.D. degree in information and communication engineering with Beijing Institute of Technology, Beijing, China.

His research interests include remote sensing object detection and recognition.



Yikang Sun was born in Hebei, China, in 2001. He received the Bachelor of Engineering degree from the University of Electronic Science and Technology of China (UESTC), Chengdu, Sichuan, China, in 2023. He is currently pursuing the master's degree with New Generation Electronic Information Technology, Beijing Institute of Technology, Beijing, China.

His research focuses on remote sensing object detection and recognition.



He Chen was born in Shenyang, China, in 1970. She received the Ph.D. degree in electronic engineering from Harbin Institute of Technology, Harbin, China, in 1998.

She is currently a Professor and the Dean of Beijing Institute of Technology in school of information, and she is working for System-on-Chip design, remote sensing data intelligent processing, VLSI architectures for real-time image, and signal processing.



Yin Zhuang (Member, IEEE) was born in Henan, China, in 1990. He received the B.S. degree in electronic engineering from the University of Sussex, Brighton, U.K., in 2013, and the Ph.D. degree from Beijing Institute of Technology, Beijing, China, in 2018.

From 2018 to 2020, he was a Postdoc with the School of Electronic Engineering and Computer Sciences, Peking University, Beijing. Currently, he has been a tenured Associate Professor with the School of Information and Communication Engineering, Beijing Institute of Technology. His research focuses on remote sensing object detection and recognition.



Jun Li (Fellow, IEEE) received the B.S. degree in geographic information systems from Hunan Normal University, Changsha, China, in 2004, the M.E. degree in remote sensing from Peking University, Beijing, China, in 2007, and the Ph.D. degree in electrical engineering from the Instituto de Telecomunicações, Instituto Superior Técnico (IST), Universidade Técnica de Lisboa, Lisbon, Portugal, in 2011.

She is currently a Full Professor with China University of Geosciences, Wuhan, China. Her main research interests include remotely sensed hyperspectral image analysis, signal processing, supervised/semisupervised learning, and active learning.

Prof. Li is the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. She has been a Guest Editor of several journals, including the Proceedings of the IEEE and the *ISPRS Journal of Photogrammetry and Remote Sensing*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:00:18 UTC from IEEE Xplore. Restrictions apply.